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General Comments : Single HW01 submission document consolidating the Problem Description, the Solution
Outline, and the conceptual Domain Model of the CSE305 Library Management System.
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HW01 — Problem Description, Solution Outline & Domain Model

Library Management System (LMS)

Course	CSE305 Software Engineering — Adnan Menderes University (ADU), Department of Computer Engineering
Instructor	Assoc. Prof. Dr. Fatih Soygazi
Project	Library Management System (LMS)
Team & Scrum roles	**Ibrahim Temur** (`231805033`) — Product Owner · **Omer Faruk Kuş** (`231805026`) — Scrum Master (both: Developer, Architect, QA, Reviewer)
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Archive	`cse305_hw01_231805033_231805026.zip` (created with `tools\package 01`)
GitHub repository	`https://github.com/ibrahimtemur/cse305-library-management-system`

> **Note**
> **How to read this document.** Parts 1 and 2 are concise consolidations. Details live in the companion documents `PROJECT_PROPOSAL.md` and `SOLUTION_OUTLINE.md` (both version `1.2`), and in the Inception documents `BUSINESS_RULES.md` (`BR-001` .. `BR-018`), `DOMAIN_GLOSSARY.md` , `REQUIREMENTS.md` (`FR-001` .. `FR-032`), and `USE_CASES.md` (`UC-01` .. `UC-11`). Part 3 (Domain Model) is specified in full here. The seed data is documented in `data/README.md` .

Contents: Part 1 — Problem Description · Part 2 — Solution Outline · Part 3 — Domain Model · Revision History

Part 1 — Problem Description

1.1 Problem Statement

University, faculty, and departmental libraries that still use paper ledgers or ad-hoc spreadsheets lose time and accuracy at the circulation desk:

1. **Manual tracking errors** — checkouts, due dates, renewals, and returns are recorded by hand (mistyped ISBNs, duplicate members, overwritten return dates, books borrowed on someone else's behalf).
2. **Untracked physical copies** — several copies of one title cannot be told apart; copies that are loaned, held for a reservation, under repair, or lost look like shelf stock.
3. **Inconsistent overdue and fine handling** — member-type quotas, overdue days, fines, and borrowing blocks are applied inconsistently.
4. **No real-time visibility** — borrowers cannot check shelf availability, their own due dates and fines, or join a fair waiting queue for a high-demand title.

1.2 Motivation

The LMS is a self-contained, offline-capable web application that automates catalog management, copy-level circulation, FIFO reservation queues, and deterministic fine calculation under the **confirmed lending rules of the ADU Library** (verified on 2026-10-22, see `BUSINESS_RULES.md`). For CSE305 the domain is a realistic vehicle for Agile Scrum, UP-style modeling, GoF/GRASP patterns, UI-first prototyping, and TDD.

1.3 Stakeholders and Roles

Role (`BR-001`)	Who & representative seeded personas (`PERSONAS.md`)	Main goals
Librarian / Admin (`ROLE_LIBRARIAN`)	Circulation staff and administrator (one unified role, `ASSUMPTION-01`; e.g., `Enginiz Yıldırım` `syn-lib-0001`)	Maintain catalog and copies, manage members, run checkout/return/renewal, settle fines, process lost-book petitions, monitor the dashboard (`UC-03`..`UC-07`, `UC-09`, `UC-11`)
Member (`ROLE_MEMBER`)	`STUDENT` (`Tutkucan Ergül` `SYN-STU-0001`), `ACADEMIC_STAFF` (`Atasagun Öcalan` `SYN-STF-0003`), or `ADMINISTRATIVE_STAFF` (`Doğannur Güçlü` `SYN-ADM-0001`)	Search the catalog, see availability, track own loans/fines, renew, reserve (`UC-01`, `UC-07`, `UC-08`, `UC-10`)
Guest (`ROLE_GUEST`)	Unauthenticated visitor	Read-only catalog search and availability (`UC-01`)
Course instructor / Scrum team	Evaluator (`Assoc. Prof. Dr. Fatih Soygazi`) / PO (`Ibrahim Temur`) and SM (`Omer Faruk Kuş`)	Traceability from vision to tests; sustainable velocity (`VISION_AND_CONTEXT.md`)

Confirmed borrowing parameters per member type (`BR-002`, `BR-003`):

Member type	Max simultaneous `ACTIVE` loans	Loan period
`ACADEMIC_STAFF`	7	30 calendar days
`ADMINISTRATIVE_STAFF`	3	30 calendar days
`STUDENT`	3	30 calendar days

Other confirmed rules in force: overdue fine `0.25 TRY` (25 kuruş) per overdue day per loan with no cap and no grace period (`BR-007`..`BR-009`); no new loan or renewal while a loan is overdue or a fine is unpaid (`BR-011`); one renewal of +30 days if not overdue and not reserved (`BR-004`, `BR-005`); 30-day re-borrowing cooldown after returning a renewed title (`BR-006`); FIFO title-level reservations with a 3-day pickup hold (`BR-012`, `BR-013`); non-circulating reference works (`BR-014`); lost-book petition freezes fines, 30-day replacement window, then twice the appraised value (`BR-015`); personal ID card, no proxy borrowing (`BR-016`).

1.4 Scope

In scope	Out of scope
Bibliographic catalog (books only, including `referenceOnly` works) seeded from Open Library	Periodicals and unpublished theses (`CON-002`)
Copy-level inventory with lifecycle states (`BR-017`)	Live ADU systems: OBIS/OBS, LDAP, SSO, smart-card readers (`CON-003`)
Member management for three member types	Payment gateways, banking APIs, SMTP/SMS (`CON-004`); fines are settled at the desk, hold notices are in-app
Checkout, return, renewal, overdue detection, fines, waivers, lost-book handling	RFID gates and scanner drivers (keyboard-emulated barcode input is supported)
FIFO reservation queue with pickup hold	E-book hosting and DRM
Role-based server-side web UI (Spring MVC + Thymeleaf), designed first in Stitch MCP	Multi-branch inter-library shipping

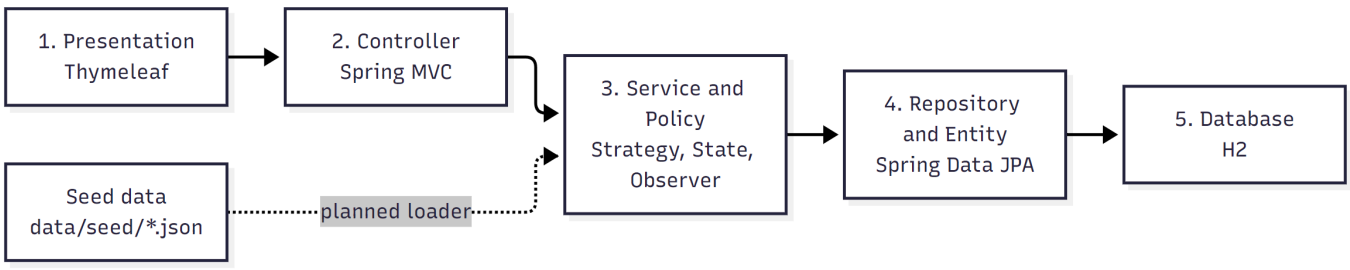
1.5 Prioritized Features (MoSCoW)

Priority	ID	Feature	FR-*	UC-*
Must	M1	Catalog & book copy management	FR-001 .. FR-005	UC-01, UC-03
Must	M2	Member management (3 member types)	FR-006 .. FR-008	UC-04
Must	M3	Loan checkout & return	FR-009 .. FR-013	UC-05, UC-06
Must	M4	Overdue detection & fine calculation	FR-014, FR-015	UC-06, UC-09, UC-11
Must	M5	Authentication, role-based views, self-service	FR-016 .. FR-018	UC-02, UC-10
Must	M6	Reproducible seed data pipeline & startup loader	FR-019	startup
Should	S1	Reservation (hold) queue & pickup hold	FR-020 .. FR-023	UC-08
Should	S2	Loan renewal	FR-024	UC-07
Should	S3	Fine settlement, waiver, lost-copy handling	FR-025 .. FR-027	UC-09
Should	S4	Librarian dashboard & overdue report	FR-028, FR-029	UC-11
Could	C1 .. C3	Guest portal; live ISBN lookup; CSV export	FR-030 .. FR-032	UC-01, UC-03, UC-11
Won't	W1	Online payments, external messaging, inter-library shipping	—	—

Part 2 — Solution Outline

2.1 Architecture

A five-tier layered web architecture (NFR-010): **Thymeleaf UI → Controller → Service/Policy → Repository/Entity → H2**. Controllers only coordinate HTTP requests (GRASP Controller); transactional services and policy/pattern classes hold the business rules (NFR-003); repositories map the persistent domain concepts to H2 (file mode at runtime, in-memory mode in tests). Details and the full diagram: SOLUTION_OUTLINE.md §1.



2.2 Technology Stack

Area	Choice	Status
Language / framework	Java 17 target (built on JDK 25 LTS), Spring Boot 3.5.16	in pom.xml
Web / persistence	Spring MVC + Thymeleaf, Spring Data JPA / Hibernate, Jakarta Bean Validation, H2	in pom.xml
Security	Spring Security with BCrypt (FR-016, FR-017, NFR-007, NFR-008)	**planned**, not yet in pom.xml
Build / tests	Maven 3.9.16 + wrapper (.\mvnw.cmd), JUnit 5, Mockito, Spring Boot Test / MockMvc	working
Seed tooling	Python 3.11 scripts in tools/data/ (Open Library fetcher, Faker tr_TR), 80 offline unittest checks	working
UI design	Stitch MCP	working

2.3 Data Strategy

Real bibliographic metadata plus 100% synthetic operational data, versioned under `data/seed/` so that startup and tests run offline (`NFR-004`, `NFR-009`, `RISK-03`). Seed v2 (reference date `2026-10-22`) contains:

File	Rows	Content
`books-openlibrary.json`	782	Open Library books, 3 of them `referenceOnly`
`members.json`	60	40 `STUDENT`, 10 `ACADEMIC_STAFF`, 10 `ADMINISTRATIVE_STAFF`
`librarians.json`	3	`ROLE_LIBRARIAN` accounts, no credentials
`copies.json`	1,706	1 to 4 copies per title
`loans.json`	164	59 `ACTIVE`, 100 `RETURNED`, 5 `LOST`
`fines.json`	47	26 `UNPAID`, 16 `PAID`, 5 `WAIVED`
`reservations.json`	22	19 `WAITING`, 3 `READY`, across 12 titles

Conventions: members `SYN-STU-####`, `SYN-STF-####`, `SYN-ADM-####`; librarians `syn-lib-####`; e-mail domain `synthetic.lms.local`; barcodes `LMS-COPY-####`; UTF-8, LF, JSON arrays, ISO-8601 dates, natural-key references, fine `amount` as a two-decimal string. No passwords, real personal data, or book prices exist in the seed files (`BR-018`, `ASSUMPTION-15`); demo credentials are BCrypt-hashed at startup (`ASSUMPTION-16`). Full format description: `data/README.md`.

2.4 UI-First Workflow

Requirements and personas → screen design in **Stitch MCP** → HTML prototype and screenshot saved under `docs/03-design/ui/` → Thymeleaf template and Spring MVC controller → MockMvc and browser verification (`AGENTS.md` §6, `NFR-001`). Eight core screens are planned (login, catalog and book detail, circulation desk, catalog/copy management, member management, self-service portal, fines and lost-book desk, dashboard); see `SOLUTION_OUTLINE.md` §3.

2.5 Design Patterns (design level, elaborated in HW03)

Pattern	Where	Why
Strategy	`LoanPolicyStrategy`, `FineCalculationStrategy`	Quotas (7/3/3), loan period, and fine tariff are policy parameters that may change (protected variations); simplified if HW03 shows a parameter table suffices
State	`BookCopyState`	Strict copy lifecycle (`BR-017`, `FR-005`)
Observer	`BookReturnedEvent` and listeners	Return triggers hold promotion and notification without coupling services (`FR-022`)
Factory / Builder	`LoanFactory`, test builders	Valid aggregates by construction
Adapter / Facade	`OpenLibraryCatalogAdapter`	Keeps third-party JSON out of entities and controllers
GRASP	Information Expert, Creator, Controller, Low Coupling, High Cohesion, Protected Variations	Responsibility assignment

None of these design-level classes appear in the Domain Model of Part 3.

2.6 Agentic Development Approach

AI agents draft documents, prototypes, and test/code scaffolding under the five engineering hats and the `Plan → Execute → Record` cadence of `AGENTS.md`; humans own scope, priorities, business rules, acceptance, and every merge into `main`. Quality gates: plan approval, automated tests, a cross-model reviewer pass, and honest audit logging in `docs/process/ai-usage-log.md` (`Pending human review` unless a named person confirmed the review). Details: `PROJECT_PROPOSAL.md` §6.3.

2.7 Test Strategy

Level	Tooling	Scope
Domain and pattern unit tests	JUnit 5, no Spring context	Quotas 3 and 7, exact due date (0 overdue days) versus next day (1), `0.25 TRY/day` without cap, 30-day cooldown boundary, fine freeze on `lostReportedOn`, copy state transitions
Repository slices	`@DataJpaTest`, in-memory H2	Mappings and unique keys (`isbn13`, `barcode`, `memberId`, `username`)
Service / web integration	`@SpringBootTest`, `@WebMvcTest`, MockMvc	Checkout, return with hold promotion, blocks, role-based access
Seed data	Python `unittest` (80 checks)	Counts, foreign keys, rule invariants, reproducibility

Gate: `.\mvnw.cmd -q test` and `tools\data\.venv\Scripts\python.exe -m unittest discover -s tools\data\tests` must pass before every merge.

Part 3 — Domain Model

The domain model follows Larman, *Applying UML and Patterns* (3rd Ed., domain model chapter): it shows **conceptual classes** of the problem domain with attributes, named associations, and multiplicities. It deliberately contains **no** controllers, services, repositories, DTOs, strategies, or state classes — those are design concepts (HW03). Attribute names are the field names of `data/seed/*.json` and `DOMAIN_GLOSSARY.md`. **Foreign-key fields** (`isbn13` in a copy or reservation, `memberId`, `copyBarcode`, `loanId`, `readyCopyBarcode`, `settledBy`) are **not** shown as attributes; the corresponding **associations** express them (see the association table).

3.1 Class Diagram

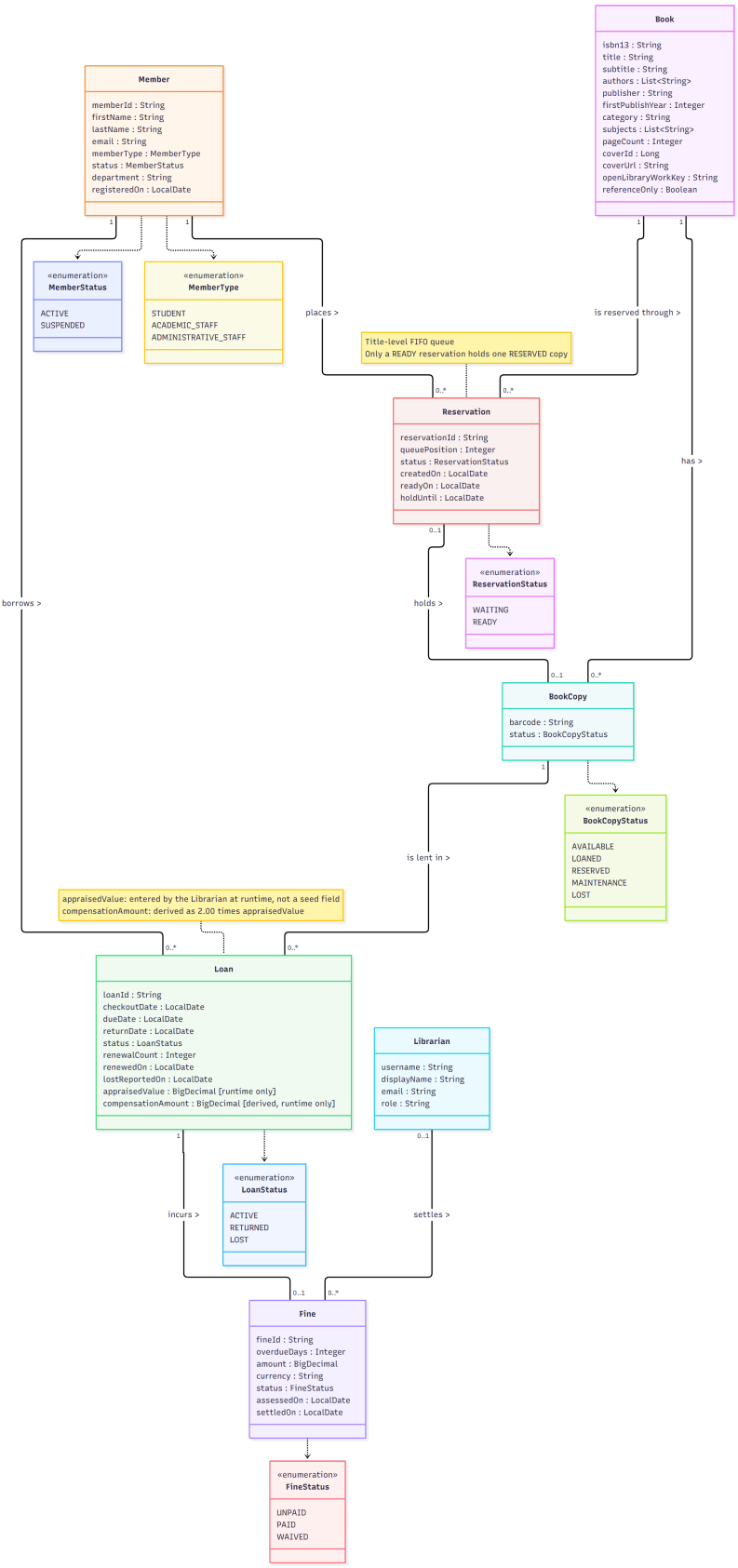


Figure 1 — Conceptual class diagram including the seven conceptual classes and six domain enumerations (vector version: `images/domain-model.svg`; source: `images/domain-model.mmd`). Attribute types follow the seed formats; the six enumeration types are also summarized with seed distributions in §3.2.

▶ Mermaid source (identical to `images/domain-model.mmd`)

3.2 Enumerations

Enumeration	Used by	Values	Seed distribution
`MemberType`	`Member.memberType`	`STUDENT`, `ACADEMIC_STAFF`, `ADMINISTRATIVE_STAFF`	40 / 10 / 10
`MemberStatus`	`Member.status`	`ACTIVE`, `SUSPENDED`	55 / 5
`BookCopyStatus`	`BookCopy.status`	`AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`	1,624 / 59 / 3 / 15 / 5
`LoanStatus`	`Loan.status`	`ACTIVE`, `RETURNED`, `LOST`	59 / 100 / 5
`FineStatus`	`Fine.status`	`UNPAID`, `PAID`, `WAIVED`	26 / 16 / 5
`ReservationStatus`	`Reservation.status`	`WAITING`, `READY`	19 / 3

`Librarian.role` is the constant string `ROLE_LIBRARIAN` (a single value, hence not an enumeration). `Guest` and the `ROLE_*` authorities are actors/roles of the system, not persistent conceptual classes (`BR-001`).

Domain parameters by member type. `memberType` is an enumeration, not a class hierarchy, because members differ only in rule parameters, not in attributes or associations (Larman: subclass only when attributes, associations, or behavior differ). The parameters are values of rules `BR-002` and `BR-003`, not model elements:

`MemberType`	Max `ACTIVE` loans	Loan period
`ACADEMIC_STAFF`	7	30 days
`ADMINISTRATIVE_STAFF`	3	30 days
`STUDENT`	3	30 days

3.3 Conceptual Classes, Seed Files, and Traceability

Conceptual class	Meaning in the domain	Seed file	Related `FR-*`	Related `BR-*`	Related `UC-*`
`Book`	A bibliographic title identified by a checksum-valid `isbn13`; may be `referenceOnly` (non-circulating)	`books-openlibrary.json`	`FR-001`, `FR-002`, `FR-003`, `FR-031`	`BR-014`, `BR-017`	`UC-01`, `UC-03`
`BookCopy`	One physical item of a title, identified by its `barcode`; has a lifecycle `status`	`copies.json`	`FR-003`, `FR-004`, `FR-005`, `FR-010`, `FR-012`	`BR-013`, `BR-014`, `BR-017`	`UC-03`, `UC-05`, `UC-06`
`Member`	A registered borrower (student, academic staff, or administrative staff)	`members.json`	`FR-006`, `FR-007`, `FR-008`, `FR-009`	`BR-002`, `BR-003`, `BR-006`, `BR-011`, `BR-016`, `BR-018`	`UC-04`, `UC-05`, `UC-10`
`Librarian`	Library staff member who operates the desk and settles fines (unified with the administrator, `ASSUMPTION-01`)	`librarians.json`	`FR-016`, `FR-017`, `FR-025`, `FR-026`	`BR-001`, `BR-010`, `BR-018`	`UC-02`, `UC-05`, `UC-06`, `UC-09`
`Loan`	The lending of one `BookCopy` to one `Member`, from checkout until return or loss	`loans.json`	`FR-009` .. `FR-014`, `FR-024`, `FR-027`	`BR-003` .. `BR-007`, `BR-011`, `BR-013`, `BR-015`, `BR-016`, `BR-017`	`UC-05`, `UC-06`, `UC-07`, `UC-09`
`Fine`	The monetary overdue penalty of one loan, with its settlement (`PAID` or `WAIVED`)	`fines.json`	`FR-014`, `FR-015`, `FR-025`, `FR-026`, `FR-029`	`BR-007` .. `BR-011`, `BR-015`	`UC-06`, `UC-09`, `UC-11`
`Reservation`	A title-level hold in a FIFO queue; when `READY` it holds one `RESERVED` copy until `holdUntil`	`reservations.json`	`FR-020`, `FR-021`, `FR-022`, `FR-023`	`BR-012`, `BR-013`, `BR-014`	`UC-06`, `UC-08`, `UC-10`

3.4 Attribute Dictionary

`?` marks attributes that may be `null` in the seed. Types follow `data/README.md` §6.

Class	Attribute	Type	Meaning
`Book`	`isbn13`	String	Unique 13-digit ISBN with valid check digit
	`title`, `subtitle?`	String	Title; optional subtitle
	`authors`	List of String	1 to 5 author names
	`publisher`	String	Publisher of the selected edition
	`firstPublishYear`	Integer	First publication year of the work
	`category`	String	One of 11 catalog categories
	`subjects`	List of String	Up to 5 raw Open Library subjects
	`pageCount`	Integer	Work-level median page count (may differ from the specific edition)
	`coverId`, `coverUrl`	Long, String	Open Library cover reference (image not stored)
	`openLibraryWorkKey`	String	Open Library work key
	`referenceOnly`	Boolean	`true` for non-circulating encyclopedias, dictionaries, catalogs (`BR-014`)
`BookCopy`	`barcode`	String	Unique `LMS-COPY-####`
	`status`	BookCopyStatus	Lifecycle state
`Member`	`memberId`	String	Unique `SYN-STU-####`, `SYN-STF-####`, or `SYN-ADM-####`; stands for the personal ID card (`BR-016`)
	`firstName`, `lastName`	String	Synthetic name
	`email`	String	`*@synthetic.lms.local`
	`memberType`, `status`	MemberType, MemberStatus	Member type and membership state
	`department`	String	Generic fictional department label
	`registeredOn`	LocalDate	Registration date
`Librarian`	`username`	String	Unique `syn-lib-####`
	`displayName`, `email`	String	Synthetic name and e-mail
	`role`	String	Constant `ROLE_LIBRARIAN`; no credential field in the seed
`Loan`	`loanId`	String	Unique `LOAN-####`
	`checkoutDate`, `dueDate`	LocalDate	`dueDate = checkoutDate + 30 * (1 + renewalCount)` days
	`returnDate?`	LocalDate	Set only for `RETURNED` loans
	`status`	LoanStatus	`ACTIVE`, `RETURNED`, or `LOST`
	`renewalCount`	Integer	`0` or `1`
	`renewedOn?`	LocalDate	Date of the renewal
	`lostReportedOn?`	LocalDate	Lost-book petition date; freezes fine accrual (`BR-015`)
	`appraisedValue` *(runtime only)*	BigDecimal	Valuation Commission value entered by the Librarian; **not in the seed** (`ASSUMPTION-15`)
	`compensationAmount` *(derived, runtime only)*	BigDecimal	`2.00 * appraisedValue`; **not in the seed**

Class	Attribute	Type	Meaning
`Fine`	`fineId`	String	Unique `FINE-####`
	`overdueDays`	Integer	Overdue days up to the effective end date (`BR-007`)
	`amount` , `currency`	BigDecimal, String	`0.25 * overdueDays` , no cap; `TRY`
	`status`	FineStatus	`UNPAID` , `PAID` , or `WAIVED`
	`assessedOn`	LocalDate	Return date, lost-petition date, or evaluation date
	`settledOn?`	LocalDate	Settlement date; `null` while `UNPAID`
`Reservation`	`reservationId`	String	Unique `RES-####`
	`queuePosition`	Integer	FIFO position `1..n` within the title's queue
	`status`	ReservationStatus	`WAITING` or `READY`
	`createdOn`	LocalDate	Placement date (FIFO order)
	`readyOn?` , `holdUntil?`	LocalDate	Hold start and `readyOn + 3 days` ; only for `READY`

3.5 Associations and Multiplicities

#	Association (read direction)	Multiplicity	Seed foreign key	Rule / invariant
A1	`Book` has `BookCopy`	1 — `0..*`	`copies.isbn13`	A title is cataloged before copies are added (`FR-001` , `FR-004`); seed has 1 to 4 per title (`ASSUMPTION-09` , `BR-017`)
A2	`Member` borrows `Loan`	1 — `0..*`	`loans.memberId`	Active-loan quota per member type (`BR-002`); blocks (`BR-011`)
A3	`BookCopy` is lent in `Loan`	1 — `0..*`	`loans.copyBarcode`	At most one `ACTIVE` loan per copy at any time (`BR-017`)
A4	`Loan` incurs `Fine`	1 — `0..1`	`fines.loanId` (unique)	At most one overdue fine per loan (`BR-010`); every fine belongs to exactly one loan
A5	`Librarian` settles `Fine`	`0..1` — `0..*`	`fines.settledBy`	`null` while `UNPAID` ; set when `PAID` or `WAIVED` (`BR-010`)
A6	`Member` places `Reservation`	1 — `0..*`	`reservations.memberId`	Eligibility (`BR-012`)
A7	`Book` is reserved through `Reservation`	1 — `0..*`	`reservations.isbn13`	Title-level FIFO queue (`BR-012`); never on a `referenceOnly` book (`BR-014`)
A8	`Reservation` holds `BookCopy`	`0..1` — `0..1`	`reservations.readyCopyBarcode`	Only a `READY` reservation holds a copy, and a `RESERVED` copy is held by exactly one `READY` reservation (`BR-013` , `BR-017`)

Derived association (not drawn). `fines.memberId` duplicates `Loan.member` (verified for all 47 seed fines). Larman advises against redundant associations, so a `Member` 's fines are reached through `Member → Loan → Fine` .

Cross-association invariants (enforced in design and tests, shown here for completeness): `BookCopy.status = LOANED` ⇔ the copy has an `ACTIVE` loan; `BookCopy.status = LOST` ⇔ the copy has a `LOST` loan (`returnDate = null` , `lostReportedOn ≠ null`); `BookCopy.status = RESERVED` ⇔ a `READY` reservation holds it; all copies of a `referenceOnly` book stay `AVAILABLE` (`BR-014` , `BR-017`).

3.6 Key Modelling Decisions

1. **`Book` versus `BookCopy` .** A title (`isbn13`) and a physical item (`barcode`) have different identities, attributes, and lifecycles: availability, loans, loss, and maintenance concern copies; catalog data, reference status, and reservations

concern the title. Merging them would make multi-copy lending and per-copy state impossible.

2. **`Member` with an enumerated `memberType`, no subclasses.** The three member types differ only in rule parameters (`BR-002`, `BR-003`), so an enumeration is enough; the parameters belong to the rules, not to the model.
3. **`Librarian` is a separate class.** Librarians and members are distinct people with different data (`librarians.json` versus `members.json`), roles, and responsibilities; a librarian appears in the domain as the person who settles a `Fine`.
4. **`Fine` is its own concept.** A fine has its own lifecycle (`UNPAID` → `PAID` or `WAIVED`), amount, assessment and settlement dates, and an accountable librarian; at most one overdue fine exists per loan (`BR-010`), and it blocks borrowing while unpaid (`BR-011`). Folding it into `Loan` would hide this lifecycle.
5. **Title-level FIFO reservations.** A `Reservation` is placed on a `Book`, not on a copy (`BR-012`); `queuePosition` and `createdOn` give the FIFO order. Only when a copy is returned does a `READY` reservation hold one specific `BookCopy` (`readyCopyBarcode`, `BR-013`).
6. **Why a lost loan has its own status.** `LOST` is a closed outcome that differs from `RETURNED`: the copy never came back (`returnDate = null`), fine accrual stopped on `lostReportedOn`, and the copy becomes `LOST`. Overloading `RETURNED` would falsely free the copy, break the invariant "`LOST` copy ⇔ `LOST` loan", and lose the fine-freeze date (`BR-015`, `BR-017`).
7. **Why `appraisedValue` is runtime-only.** Open Library data contains no prices, and inventing fake prices in the seed would be false data. The Valuation Commission's value is entered by the Librarian when a lost copy is not replaced within 30 days (`ASSUMPTION-15`); `compensationAmount` is derived as `2.00 * appraisedValue`. Both appear on `Loan` as attributes marked *runtime only* and are not seed fields.
8. **Associations instead of foreign keys.** Following Larman, relationships are drawn as named associations with multiplicities; the seed's natural-key fields are the implementation of those associations and are mapped in §3.5. The redundant `Fine` – `Member` link is omitted.
9. **`Reservation` states `WAITING` and `READY` only.** This equals the glossary and the seed. A reservation that is fulfilled, cancelled, or expired leaves the active queue (see Known Limitations).

3.7 Known Limitations

- **Lost-book compensation tracking is deferred to HW03.** The model records `appraisedValue` and the derived `compensationAmount` on `Loan` only. How the compensation charge is **tracked, paid, or waived** (for example its own status, `settledOn`, and `settledBy`, or an extension of `Fine`) is not modelled in HW01 and will be decided in the HW03 design documents. Until then, `FR-025` / `FR-026` (settlement and waiver) are fully modelled for overdue `Fine` records only.
- **Reservation terminal states and history are deferred to HW03.** `ReservationStatus` contains only `WAITING` and `READY`. `FR-011` (fulfilled by checkout) and `FR-023` (cancellation, expiry) remove a reservation from the active queue; whether terminal states such as fulfilled, cancelled, or expired, and a retained reservation history, are needed will be decided in HW03. The seed contains no such records.
- The model is conceptual: it carries no operations, visibility, or persistence mapping.
- Seed dates are relative to `REFERENCE_DATE = 2026-10-22`; shifting them at startup is open item `OPEN-04`.

3.8 Concepts Intentionally Left to the HW03 Design Documents

Design-level concept	Why it is not in the domain model
`LoanPolicyStrategy`, `FineCalculationStrategy` and their implementations	Strategy classes are design solutions for policy variation, not domain concepts
`BookCopyState` and its state classes	State pattern realizes the `BookCopyStatus` lifecycle
`BookReturnedEvent`, `LoanOverdueEvent`, listeners	Observer mechanism
`LoanFactory`, builders	Creational design
`Notification` (in-app hold-ready message, `BR-013`)	Runtime artifact; its need as a persistent concept is a design decision
User account / credentials / security roles	Login data is created at startup (`ASSUMPTION-16`); not in the seed

Design-level concept	Why it is not in the domain model
Controllers, services, repositories, DTOs, JPA entity mappings	Layered-architecture elements
Compensation charge tracking; reservation history	See Known Limitations

Revision History

Version	Date	Summary
`1.2`	2026-10-30	First consolidated HW01 submission document: Problem Description, Solution Outline, and Domain Model; aligned with confirmed ADU rules and seed v2; schedule updated (HW01 due 2026-11-03).